ARTIFICIAL INTELLIGENCE

Chapter 2
Intelligent Agents

Agenda



Environment



Agent Frameworks



QUIZ

AI Program is called?

- Wetware
- Formula
- Intelligent Agent

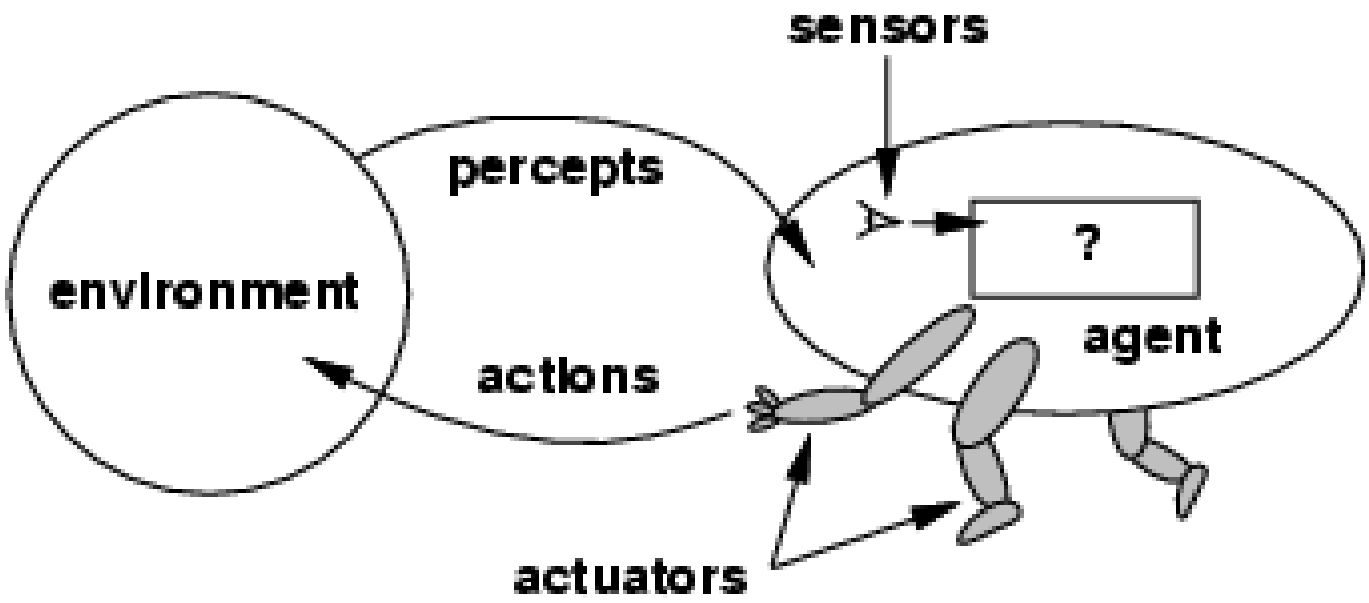
Definition of Agent

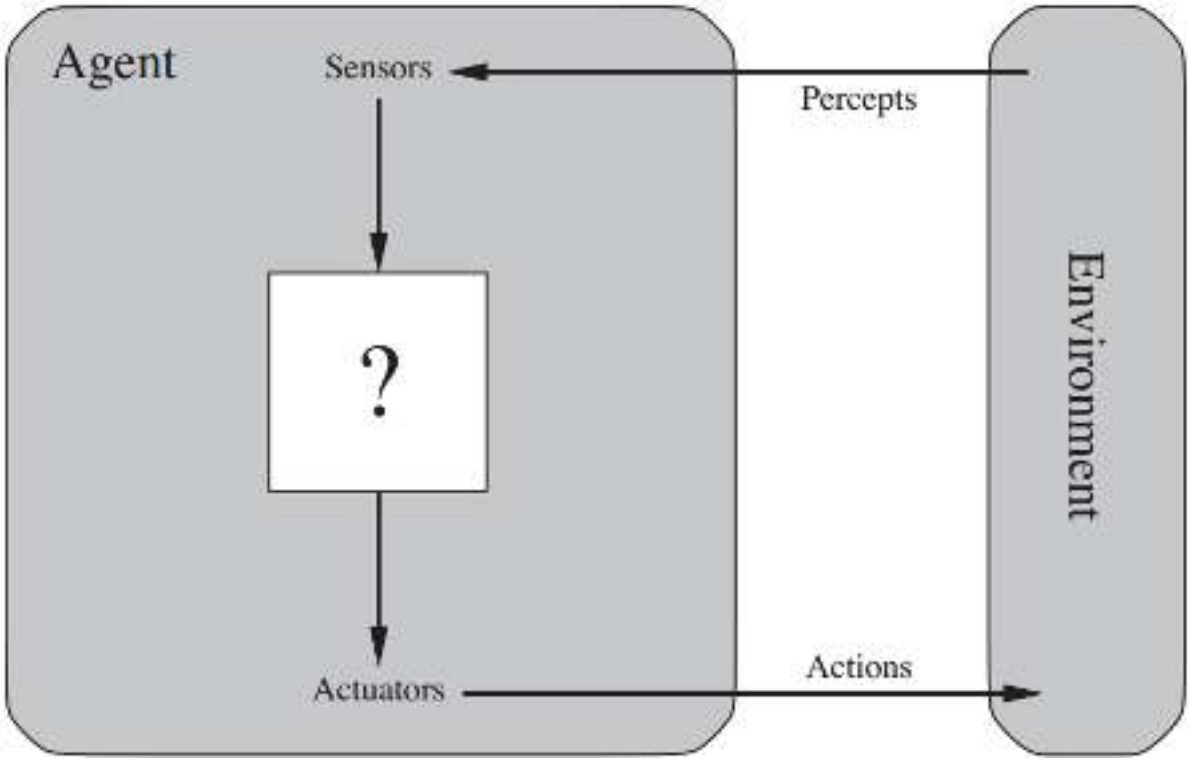
Anything that:

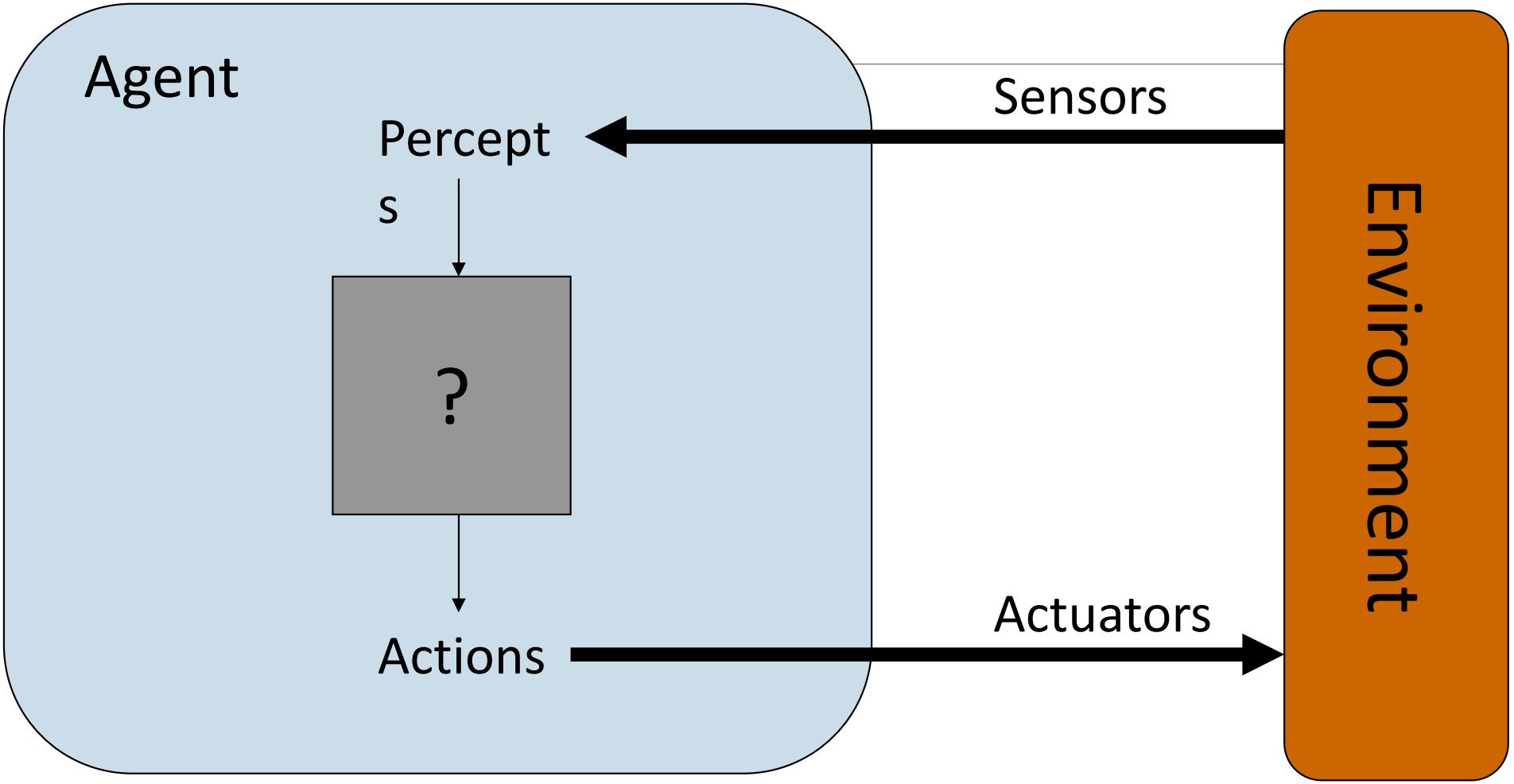
- **Perceives** its environment
- **Acts** upon its environment
- E.g. controller, robot



- An **agent** is anything that can perceive its **environment** through sensors and acts upon that **environment** through effectors.


sensors
percepts
environment
actions
?
agent
actuators


Agent
Sensors
Percepts
?
Actuators
Actions
Environment



Sense – Plan – Act

Definition of "Environment"

The real world, or a virtual world

- Rules of math/formal logic
- Rules of a game

…

- Specific to the **problem domain**

What is a rational Agent?

A **rational agent** is one that does the right thing.

- In order to know the right thing, we need a performance measure.
- The **performance measure** is usually chosen by the agent designer.
- For each possible percept sequence, a rational agent should select an **action** that is expected to maximize the performance measure given the evidence provided by the percept sequence and whatever built-in **knowledge** the agent has.

Specifying the Task Environment

Task Environments are “problems” to which rational agents are the solutions!

To specify a task environment, four items need to be defined:

- **P**erformance measure
- **E**nvironment
- **A**ctuators
- **S**ensors

→ This is **PEAS** description

Specifying the Task Environment

Example: PEAS description of the task environment for an automated taxi.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi Driver	Safe, fast, legal, maximize profits, …	Roads, other traffic, pedestrians, customers, …	Steering, accelerator, signal, horn, …	Cameras, sonar, speedometer, GPS, …

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts: bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, beaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises. suggestions, corrections	Keyboard entry

Figure 2.5 Examples of agent types and their PEAS descriptions.

Properties of environments

Properties of the Environment

Fully observable vs. partially observable:

- See everything vs. hidden information.

Deterministic vs. stochastic vs. strategic:

- Controlled by agent vs. randomness vs. multi-agents.

Episodic vs. sequential:

- Independent episodes vs. series of events.

Static vs. dynamic vs. semi-dynamic:

- World waits for agent vs. world goes on without agent vs. world waits but agent timed.

Discrete vs. continuous:

- Finite distinct states vs. uninterrupted sequence.

Single agent vs. cooperative vs. competitive:

- Alone vs. team-mates vs. opponents.



Fully Observable vs. Partially Observable

- ***fully observable:*** An environment is called fully observable if what your agent can sense at any point in time is completely sufficient to make the optimal decision i.e. its sensors can see the entire state of the environment. Ex. Image recognition, chess
- ***partially observable***: that is in contrast to some other environments where agents need memory to make the best decision. Ex. self-driving vehicle scenarios

Deterministic vs. Stochastic

- If the next state of the environment is completely determined by the current state and the actions of the agent, then the environment is deterministic; otherwise it is non-deterministic.
- Stochastic = the next state has some uncertainty associated with it.
- Uncertainty could come from randomness, lack of a good environment model, or lack of complete sensor coverage.

Episodic / Sequential

- In an episodic environment, each episode consists of the agent perceiving and then acting. The quality of its action depends just on the episode itself.
- Subsequent episodes do not depend on the actions in the previous episodes.
- Sequential if current decisions affect future decisions, or rely on previous ones

Static vs. Dynamic

- If the environment does not change while an agent is acting, then it is ***static***; otherwise it is dynamic. Speech analysis is a problem that operates on static AI environments.
- Contrasting with that model, ***dynamic*** AI environments such as the vision AI systems in drones deal with data sources that change quite frequently.

Discrete vs. Continuous

- ***Discrete*** AI environments are those on which a finite [although arbitrarily large] set of possibilities can drive the final outcome of the task. Chess is also classified as a discrete AI problem.

- ***Continuous:*** Signals constantly coming into sensors, actions continually changing. Vision systems in drones or self-driving cars operate on continuous AI environments.

Competitive vs. Collaborative

- ***Competitive*** AI environments face AI agents against each other in order to optimize a specific outcome. Games such as GO or Chess are examples of competitive AI environments.

- C***ollaborative*** AI environments rely on the cooperation between multiple AI agents. Self-driving vehicles or cooperating to avoid collisions or smart home sensors interactions are examples of collaborative AI environments.

Examples of environments and their properties:

Environment	Obs.	Det.	Eps.	St.	Disc.	Agent
Sudoku	Fully	Det.	Seq.	St.	Disc.	Single
Chess	Fully	Strat.	Seq.	Semi	Disc.	Comp.
Robot Soccer Players	Part.	Stoc.	Seq.	Dyn.	Cont.	Comp./ Coop.
Part-Picking Robot	Part.	Stoc.	Eps.	Dyn.	Cont.	Single

Exercise

Formulate the problem domains for:

- Tic-tac-toe
- A web server
- An insect
- A student in AI course
- A doctor diagnosing a patient
- Football team
- Saudi Arabia

What is/are the:

- PEAS

How might a "good-behaving" agent process information?

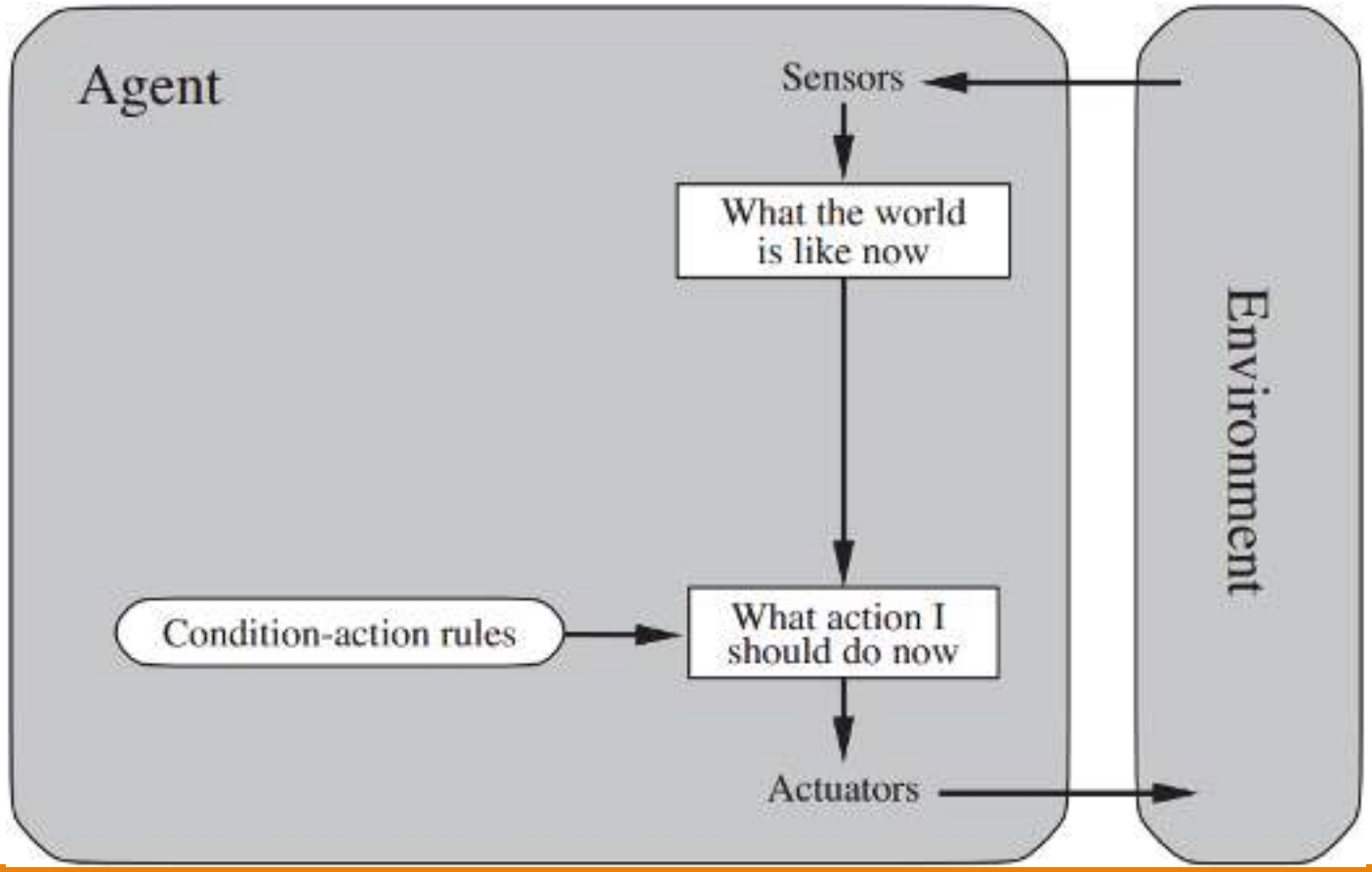
Types of agents

Types of agents

- Simple reflex (aka reactive, rule-based)
- Model-based
- Goal-based
- Utility-based (aka decision-theoretic, game-theoretic)
- Learning (aka adaptive)

Simple Reflex Agents

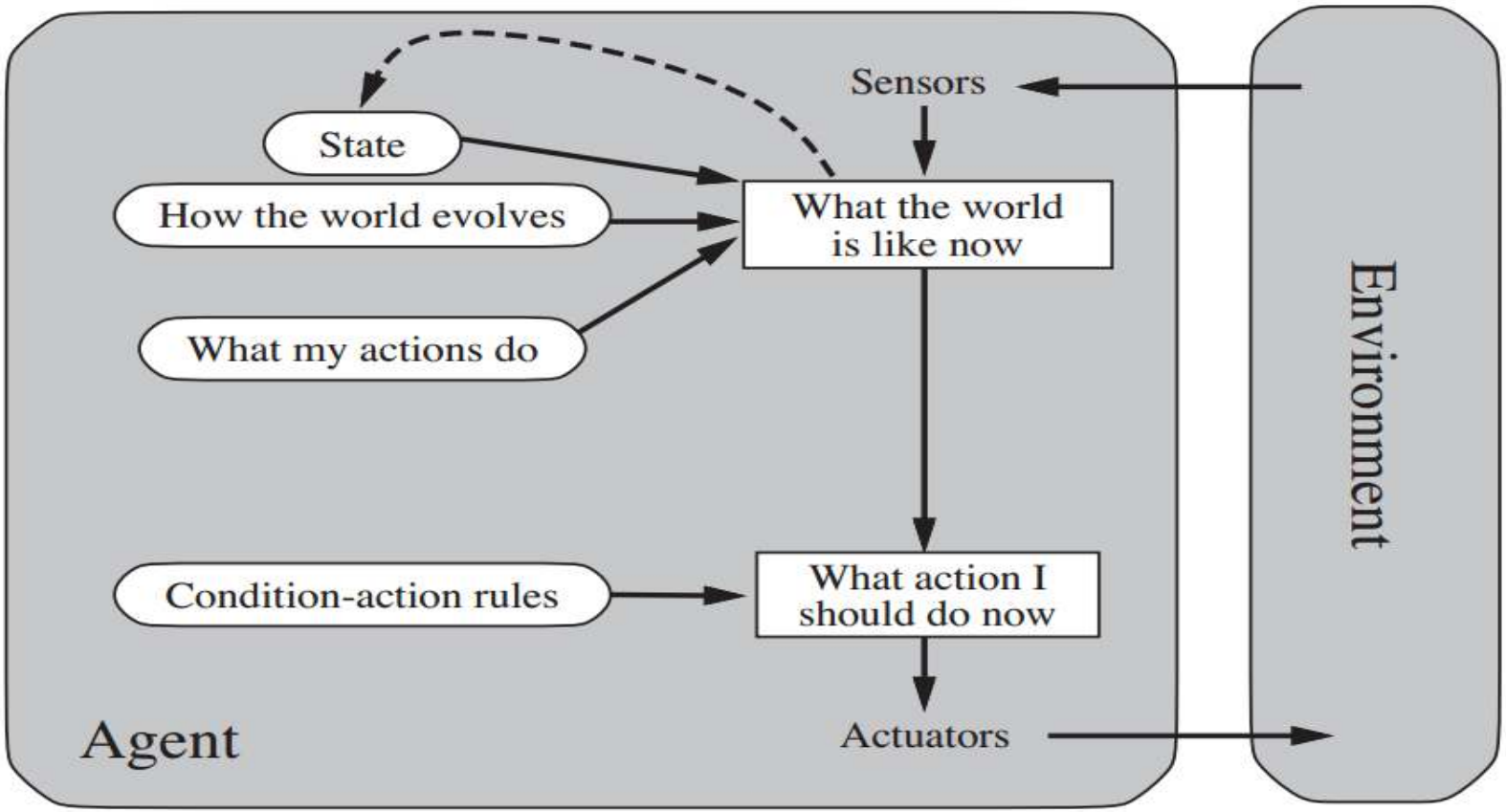
- Selects action based only on current perception of the environment.
- Simple but has a limited intelligence!



Model-Based Reflex Agent

- Keeps track of perception history in the form of internal state

(reflecting unobserved aspects of the current state of the World).

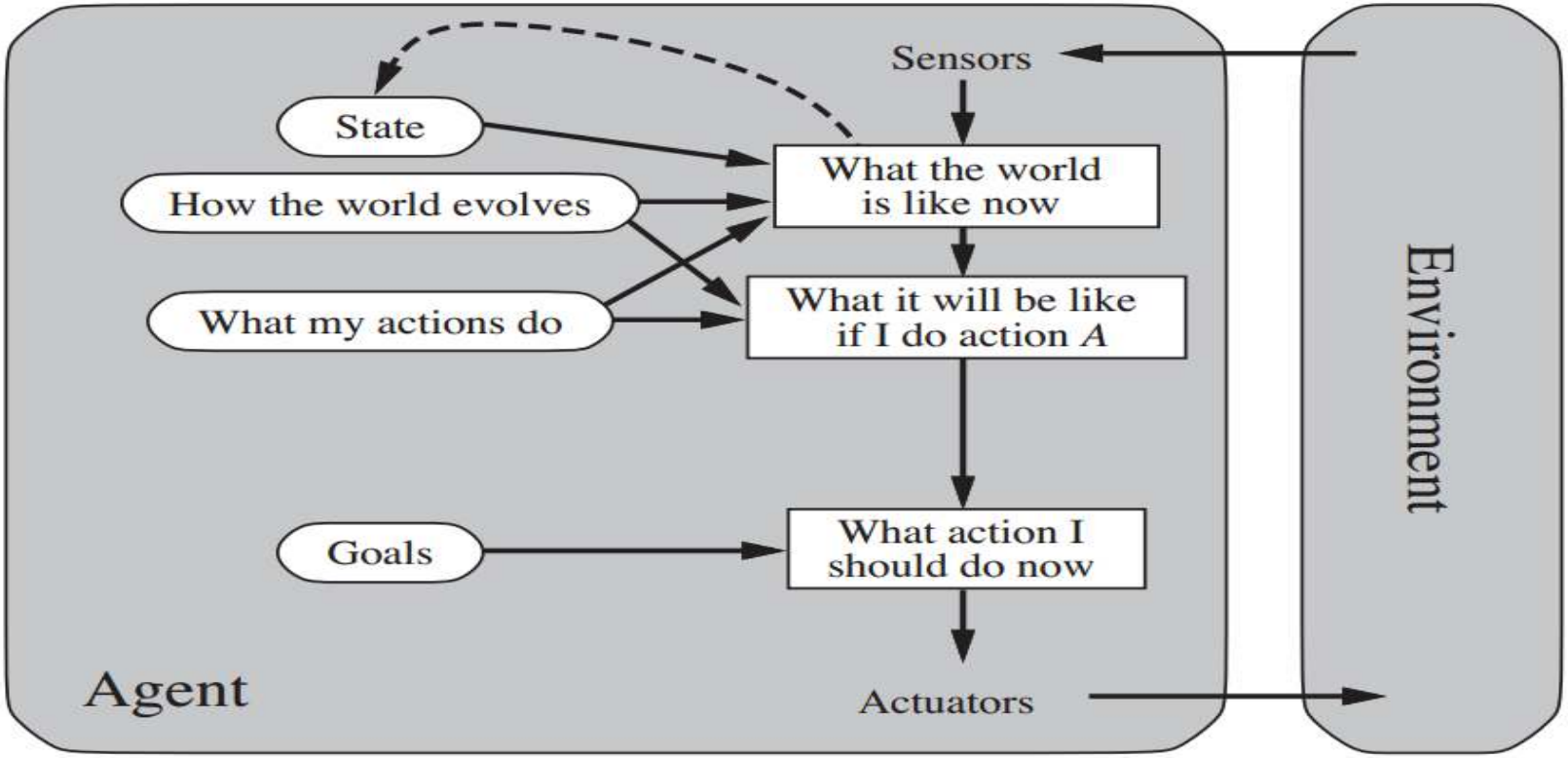


Goal-Based Agent

- Keeps track of world state and goals to be achieved.

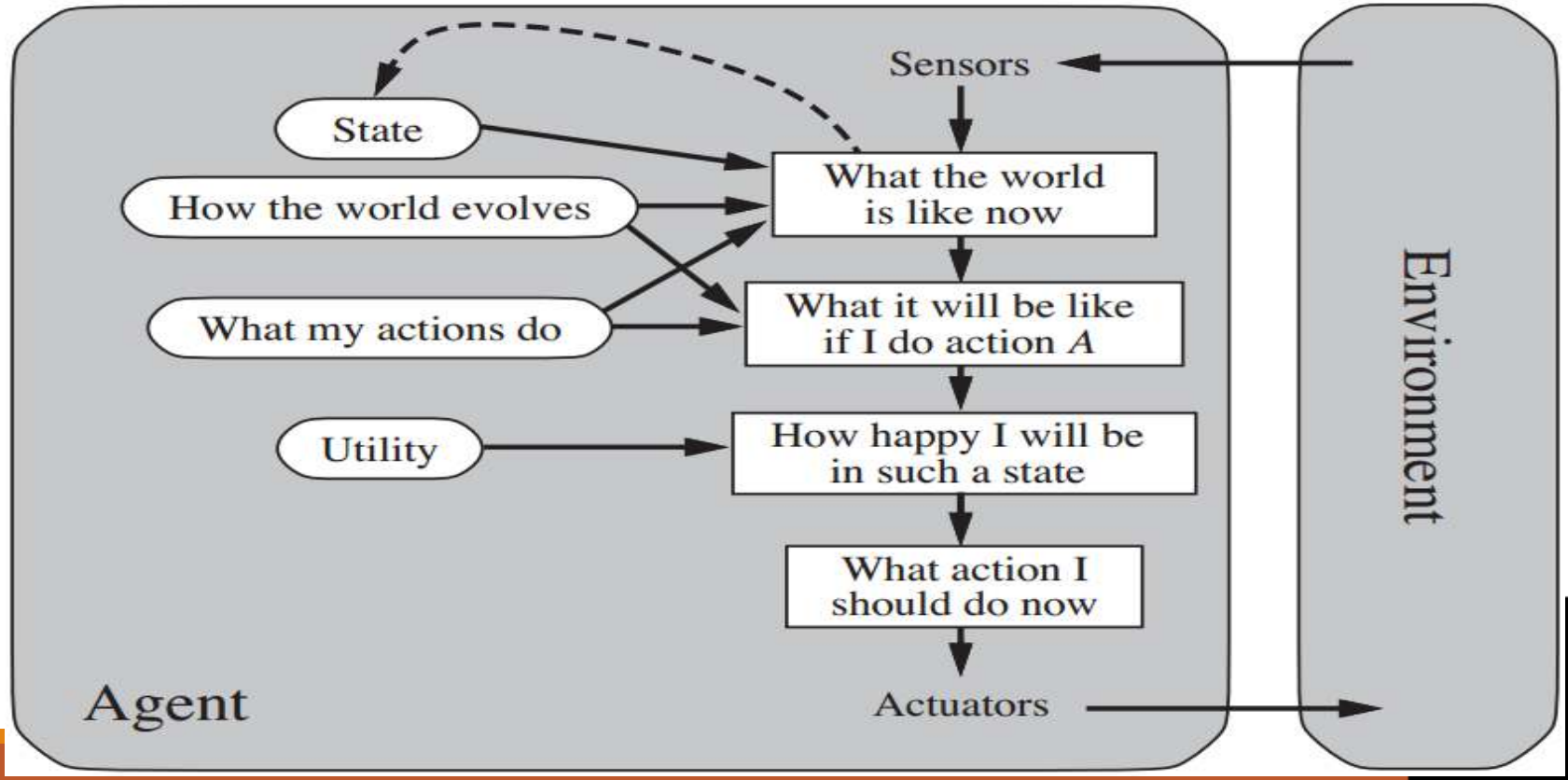
- Chooses action that will lead to achieving its goals.

Flexible (knowledge & goal can be easily modified).



Utility-Based Agent

- Keeps track of world state and uses utility function to measure performance among world states.
- Agent chooses the actions that maximize its utility.



Learning Agent

Adds the ability to learn from its experiences.

- Learning element: makes improvements; Performance element: selects external actions; Critic: determines how agent is doing; Problem generator: suggests exploratory actions.

